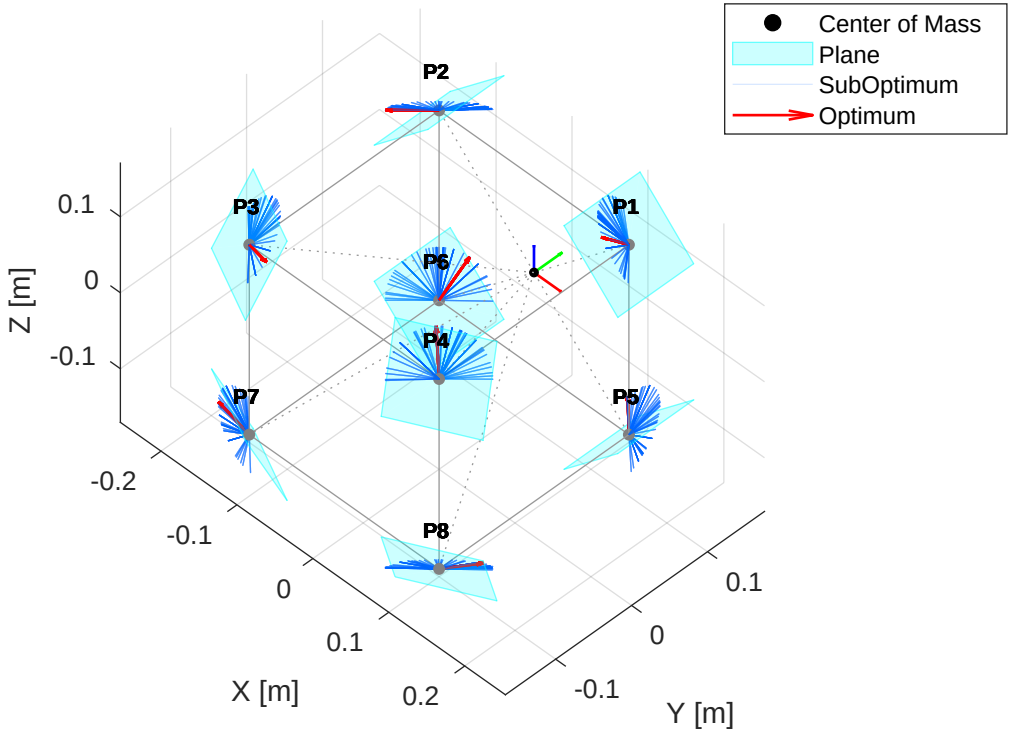CoM Shift: $\Delta x = 0.0$, $\Delta y = 0.125$, $\Delta z = 0.0$)